

SAYAM SETHI

Email  | Google Scholar  | GitHub  | LinkedIn 

EDUCATION

PhD in Electrical and Computer Engineering

August 2023 – 2028 (tentative)

Advisor: Dr. Jonathan Baker; The University of Texas at Austin

Austin, Texas

GPA: 3.88/4

Bachelor of Technology | Dept. of Computer Science and Engineering

July 2019 – May 2023

Indian Institute of Technology, Delhi

Delhi, India

GPA: 9.55/10

RESEARCH DIRECTION

I am interested in working on Quantum Error Correcting (QEC) architectures, specifically the compilation aspect for different error-correcting codes such as surface codes, qLDPC etc. I am interested in looking at dynamic compilation which involve making circuit execution decisions based on measurement results and/or Repeat-Until-Success (RUS) protocols. I am also interested in exploring problems related to minimising resource consumption and utilisation, for both classical post-processing overheads such as decoders, and quantum resources such as number of ancillas, distillation factories etc.

PUBLICATIONS

LOQUS: Logical Qubit Mapping Problem on Bivariate Bicycle Codes

S. Sethi^{*}, S. Khan^{*}, et al.

Under review

- Designed a logical qubit mapper to map logical qubits within gross code modules, and to map the modules onto the architecture
- Utilized hypergraph partitioning and a novel placement heuristic to address the mapping problem
- Achieved up to 36% reduction in inter-module errors, the most expensive gate operation in the tour de gross ISA, and an average of 13% reduction in error rates.

RESCQ: Realtime Scheduling for Continuous Angle QEC Architectures

S. Sethi, and J. Baker

30th ACM International Conference on Architectural Support for Programming Languages and OS

- Worked with a non-traditional quantum surface code architecture that uses small angle rotation gates instead of magic state distillation
- Implemented a just-in-time compilation and scheduling strategy that uses ancilla qubits in the local neighbourhood to execute different gates
- Achieved 2x improvement in execution time compared to the baseline compilation and scheduling strategies

Cyclic Qubit Mappings

M. Poster, S. Sethi, and J. Baker

2024 IEEE International Conference on Quantum Computing and Engineering (QCE)

- Investigated effects of variable error rates in a surface code grid on the program fidelity.
- Proposed a solution to achieve the average case error rate irrespective of the error distribution by cyclically mapping the data qubits over different paths of ancilla qubits.

Optimizing 3D DRAM Performance under Power and Thermal Constraints

Jan 2022 – May 2023

IEEE TCAD Volume: 43 Issue: 8

Undergraduate Thesis

Memory Level:

- Studied the memory access patterns and their affect on the temperature of the HBM2E 3D Memory for multiple SPEC and PARSEC workloads
- Leveraged the thermal correlation between the vertically adjacent ranks with a reward-based power allocation policy **3D-TemPo** to simultaneously perform Dynamic Thermal Management
- Benchmarked with multiple baseline integrated policies such as round-robin, MFU, etc and existing DTM policies to achieve an execution time speedup of upto 17.94 times

System Level: (extension to the publication)

- Extended the 3D-TemPo work to incorporate power budgeting at the system level, for both memory and cores using a modified reward function
- Greatly reduced the energy consumption and execution times by using an average of 20% lesser energy and 15% lesser time than the baseline implementation

INTERNSHIPS

IBM Research, TJ Watson Research Center

Summer 2024 and Summer 2025

Quantum Research Scientist Intern

Compilation for the Gross Code (Summer 2025):

- Optimising the compilation pipeline proposed in Tour de gross by reducing the overheads incurred in inter-module measurements and Clifford rotations inserted in single-module measurements
- Aims to reduce the noise cost incurred at each gate of circuit execution, thereby increasing the depth of the circuits that can be run on IBM Starling, and other future IBM quantum processors

Exploring IQP Circuits (Summer 2025):

- Investigating classical trainability of parameterised IQP circuits which can then be deployed on quantum systems, enabling scalable quantum generative models
- Showing separation for IQP vs classical for classically verifiable datasets inspired from quantum games

Explainability for QML (Summer 2024):

- Systematically explored the space of **Explainable AI** in the context of **Quantum Machine Learning** to understand and compare the learning process of quantum models versus classical models
- Designed a framework to compute Shapley values for an arbitrary black-box model that involves projected quantum feature maps such as Heisenberg ansatz, Pauli feature maps, etc
- Observed significantly lower correlations (upto a drop of 0.5) in the explanations given by models using quantum ansatz as compared to the explanations given by purely classical models

Quadeye Securities LLP, Gurugram

Jun – Jul 2022

Systems Intern

- Developed an HTTP/2 Client library along with a parser to encode and decode HTTP/2 frames
- Designed data structures to reduce latency of features like stream lifecycle, HPACK and Huffman encoding
- Enhanced security by adding TLS support, extensively tested library using unit tests and stress testing
- Performed optimisations to reduce i-cache misses and memory allocation to get an average parse time of 300ns

ACADEMIC ACHIEVEMENTS

- **Texas Quantum Institute Graduate Fellowship (2025-27):** Awarded a fellowship for investigating classical overheads in FTQC compilation
- **IIT Delhi Semester Merit Award:** Secured for being in the **top 7 percentile** in **Semester I** of **2019-20** and **2021-22**
- **ICPC India Regionals, Amritapuri '21:** Secured **India Rank 47** and **Institute Rank 1** out of **4393 teams**
- **Google HashCode '22:** Secured **India Rank 12**, **Institute Rank 1** and **Global Rank 240** out of **10177 teams**
- **Indian National Mathematical Olympiad:** Qualified for **INMO 2018** (precursor to IMO) on being shortlisted in the Telangana region in **RMO 2017**
- **Kishore Vaigyanik Protsahan Yojana (KVPY):** Offered the fellowship in **SX stream (2018-19)** and **SA stream (2017-18)** with **All India Ranks** of **34** and **271** respectively

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Java, VHDL, Assembly, Standard ML, Octave, MATLAB, R
Libraries/Frameworks and Tools: Qiskit, Stim, Sniper, HotSpot, CoMeT, Vivado, L^AT_EX, git

TEACHING EXPERIENCE AND EXTRA-CURRICULAR ACTIVITIES

Teaching Assistant

- Grad TA for ECE 382V - Quantum Error Correction Theory & Systems in Spring '2
- Grad TA for ECE 312 - Software Design & Implementation in Fall '24, Spring '25, course for ECE sophomores
- Grad TA for ECE 306 - Introduction to Computing in Fall '23, introductory course for ECE freshers
- Head TA for COL216 (**received the outstanding TA award**) - Computer Architecture in Spring '23, core course for CSE sophomores
- TA for COL215 - Digital Logic and System Design in Fall '22, core course for CSE sophomores

Algorithms and Coding Club, IIT Delhi

- **Overall Coordinator, 2022-23:** Got the club promoted from a Chapter to a Technical Club; organized events in Tryst, Technical Fest at IITD
- **Coordinator, 2021-22:** Conducted the Summer of CP and took live lectures on YouTube; guided executives
- **Executive, 2020-21:** Authored various problems for Competitive Programming contests conducted by the club